

SKY is the LIMIT

A tale of Limits, Infinity, and Indeterminate Forms

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Lecture Objectives

- Understand the concept of a limit intuitively and formally.
- Evaluate limits of functions, including those that approach infinity or zero.
- Distinguish between undefined and indeterminate forms.
- Apply L'Hôpital's Rule to evaluate indeterminate limits.

1 Part 1: The Foundation - What is a Limit?

1.1 Intuitive Idea

A limit describes the value that a function **approaches** as the input (x) approaches some value. It's about the trend, not necessarily the value at the point itself.

- **Notation:** $\lim_{x \rightarrow a} f(x) = L$
- **Meaning:** As x gets closer and closer to a , the function $f(x)$ gets closer and closer to L .

1.2 A Puzzling Start: Is $0.999\ldots = 1$?

This is a classic question that challenges our intuition about limits and infinite series.

- **The Answer:** Yes, $0.999\ldots$ is exactly equal to 1.
- **Why?**

1. **Algebraic Proof:** Let $x = 0.999\dots$. Then $10x = 9.999\dots$. Subtract the first equation from the second:

$$10x - x = 9.999\dots - 0.999\dots$$

$$9x = 9$$

$$x = 1$$

2. **Limit/Series Proof:** $0.999\dots$ can be written as the infinite geometric series $0.9 + 0.09 + 0.009 + \dots$. The sum of this series is:

$$\frac{a}{1-r} = \frac{0.9}{1-0.1} = \frac{0.9}{0.9} = 1$$

- **The Takeaway:** This demonstrates that an infinite process can converge to a finite, precise value—a core idea in calculus.

2 Part 2: Evaluating Basic Limits

2.1 Limits of Trigonometric Functions

Let's evaluate $\lim_{x \rightarrow \pi/2} (\sin x, \cos x, \tan x)$.

- $\lim_{x \rightarrow \pi/2} \sin x$: As x approaches $\pi/2$ (90°), $\sin x$ approaches **1**. So, the limit is **1**.
- $\lim_{x \rightarrow \pi/2} \cos x$: As x approaches $\pi/2$, $\cos x$ approaches **0**.
- $\lim_{x \rightarrow \pi/2} \tan x$: Since $\tan x = \sin x / \cos x$, and $\cos x$ is approaching 0, this function grows without bound. The limit **does not exist (DNE)** in the finite sense; it approaches infinity.

2.2 Limits at Infinity

We are interested in the behavior of a function as x becomes very large ($x \rightarrow +\infty$) or very small ($x \rightarrow -\infty$).

- **Example:** $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+1}}{2x+5}$
- **Technique:** For rational expressions (or those with roots), a powerful technique is to divide the numerator and denominator by the highest power of x in the denominator.

1. Highest power of x here is x .
2. Divide numerator and denominator by x . Remember that $\sqrt{x^2} = |x|$, and since $x \rightarrow +\infty$, $|x| = x$.

$$\lim_{x \rightarrow \infty} \frac{\frac{\sqrt{x^2+1}}{x}}{\frac{2x+5}{x}} = \lim_{x \rightarrow \infty} \frac{\sqrt{1+\frac{1}{x^2}}}{2+\frac{5}{x}}$$

3. As $x \rightarrow \infty$, $\frac{1}{x^2} \rightarrow 0$ and $\frac{5}{x} \rightarrow 0$.
4. The limit simplifies to $\frac{\sqrt{1+0}}{2+0} = \frac{1}{2}$.

3 Part 3: The Philosophical Core - Zero, Infinity, and Indeterminacy

3.1 Problems with Zero

- **1/0 is Undefined.** Division by zero is not allowed. As a limit, if the denominator approaches 0 while the numerator does not, the value of the function approaches **infinity** (or **negative infinity**). It is not a number; it's a concept of unbounded growth.

3.2 What is Infinity? (∞) - A Limit-Based Perspective

- **Infinity is a concept, not a number.** In calculus, the symbol ∞ represents unbounded growth. We use it to describe the behavior of a function or a sequence as its input or index becomes arbitrarily large.
- **"Operations with Infinity" are Shorthand for Limit Laws:** It is a critical misconception to treat infinity as a number you can plug into equations. Statements like $a/\infty = 0$ are not arithmetic rules; they are shorthand summaries of the behavior of limits. The correct way to express these ideas is through the language of limits:

– **" $a / \infty = 0$ " is a Misleading Statement.**

* **Correct Interpretation:** If $\lim_{x \rightarrow c} f(x) = L$ (where L is a finite number) and $\lim_{x \rightarrow c} g(x) \rightarrow \infty$ (or $-\infty$), then the limit of the quotient is zero.

* **Limit Law:** $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = 0$

* **Example:** $\lim_{x \rightarrow \infty} \frac{5}{x} = 0$ because the denominator grows without bound, dwarfing the finite numerator.

– **" $\infty + a = \infty$ " is a Misleading Statement.**

* **Correct Interpretation:** If $\lim_{x \rightarrow c} f(x) \rightarrow \infty$ and $\lim_{x \rightarrow c} g(x) = L$ (a finite number), then the limit of the sum is still infinity.

* **Limit Law:** $\lim_{x \rightarrow c} [f(x) + g(x)] \rightarrow \infty$

– **" $\infty \cdot \infty = \infty$ " is a Misleading Statement.**

* **Correct Interpretation:** If two functions both grow without bound, their product will also grow without bound.

* **Limit Law:** If $\lim_{x \rightarrow c} f(x) \rightarrow \infty$ and $\lim_{x \rightarrow c} g(x) \rightarrow \infty$, then $\lim_{x \rightarrow c} [f(x) \cdot g(x)] \rightarrow \infty$.

- **The Key Takeaway:** Never perform "arithmetic with infinity." Instead, understand that these phrases are intuitive summaries of the results of **limit processes**. The outcome of a limit that appears to involve infinity depends entirely on the relationship and growth rates of the functions involved, which is why indeterminate forms exist.

3.3 The Great Confusion: Undefined vs. Indeterminate

This is a critical distinction.

- **Undefined:** The operation makes no sense. It is **meaningless**. There is no possible value.

– **Examples:** $1/0$, $0/0$, $\infty - \infty$, $\log(0)$.

- **Indeterminate Form:** It is a "clash of titans." The form itself does not tell you what the limit is. The limit could be 0, ∞ , 5, or anything in between. It requires further analysis (like simplification, L'Hôpital's Rule, or series expansion).

– **The "7 Deadly Sins" (Common Indeterminate Forms):**

1. $0/0$
2. ∞/∞
3. $0 \cdot \infty$
4. $\infty - \infty$
5. 0^0
6. 1^∞
7. ∞^0

Why is $0/0$ indeterminate? Consider these limits, all of which are of the form $0/0$:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{x}{x} &= 1 \\ \lim_{x \rightarrow 0} \frac{5x}{x} &= 5 \\ \lim_{x \rightarrow 0} \frac{x^2}{x} &= 0 \\ \lim_{x \rightarrow 0} \frac{x}{x^2} &= \infty\end{aligned}$$

The same form $0/0$ gives different results! The form is indeterminate.

4 Part 4: Advanced Techniques for Evaluating Limits

4.1 The Formal (Epsilon-Delta) Definition of a Limit

- This is the rigorous mathematical definition that formalizes the intuitive idea of "getting closer."
- **Definition:** $\lim_{x \rightarrow a} f(x) = L$ if for every number $\epsilon > 0$ (no matter how small), there exists a number $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \epsilon$.
- **In simpler terms:** You can make $f(x)$ as close as you want to L by making x sufficiently close (but not equal) to a .

4.2 L'Hôpital's Rule - The Ultimate Weapon for Indeterminate Forms

This rule provides a powerful method for evaluating limits of the indeterminate forms $0/0$ or ∞/∞ .

- **The Rule:** If $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ is of the form $0/0$ or ∞/∞ , then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists or is infinite.

- **Example:** Find $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ (a classic $0/0$ form).
 1. Check: $\sin(0) = 0$, denominator 0. It's $0/0$. Apply L'Hôpital's Rule.
 2. Differentiate numerator and denominator: $f'(x) = \cos x$, $g'(x) = 1$.
 3. The new limit is $\lim_{x \rightarrow 0} \frac{\cos x}{1} = \cos(0) = 1$.
 4. $\therefore \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

BUT DO YOU KNOW THE ABOVE ONE IS NOT A VALID PROOF?

FIND OUT WHY? HINTS: Which comes first? HEN Or EGG?

Once you find it out, you'll understand why we need:

SQUEEZE/SANDWICH THEOREM

- **Use of Logarithms:** L'Hôpital's Rule is especially useful for indeterminate forms involving exponents (0^0 , 1^∞ , ∞^0).
 - **Technique:**
 1. Let $y = [f(x)]^{g(x)}$.
 2. Take the natural log of both sides: $\ln y = g(x) \cdot \ln(f(x))$. This converts the problem into a $0 \cdot \infty$ form.
 3. Rewrite as $\ln y = \frac{\ln(f(x))}{1/g(x)}$ to get a $0/0$ or ∞/∞ form.
 4. Apply L'Hôpital's Rule to find $\lim \ln y$.
 5. Since $y = e^{\ln y}$, the original limit is $e^{\lim \ln y}$.

Summary & Key Takeaways

1. **A limit** describes the behavior of a function as it *approaches* a point, not necessarily at the point.
2. **Infinity** (∞) is a concept of unbounded growth, not a number. Statements like $1/\infty$ are not arithmetic; they are shorthand for a **limit process** where a function with a finite numerator is dominated by a function whose magnitude grows without bound, resulting in a limit of zero.

3. **Crucial Distinction:**

- **Undefined:** The expression has no meaning (e.g., $1/0$).
- **Indeterminate Form:** The limit could be any value and requires deeper analysis (e.g., $0/0$, ∞/∞).

4. **L'Hôpital's Rule** is a powerful tool for evaluating $0/0$ and ∞/∞ limits by using derivatives.

BEST WISHES in your studies! Mastery of limits is the essential first step to understanding all of differential and integral calculus.